ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

NATURAL SCIENCES

PRACTICE EXAM 2

MARKS: 100 **TIME: 2 hours**

Covers all four Grade 7 strands: Life and Living (biosphere, plant and animal classification, human reproduction, variation), Matter and Materials (properties, recycling, distillation, acids and bases, metals and non-metals), Energy and Change (potential and kinetic energy, insulation investigation, renewable energy, electricity supply) and Planet Earth and Beyond.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully and answer only what is asked.
4. Write in full sentences where explanations are asked for.
5. Diagrams are NOT necessarily drawn to scale.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which of the following is NOT a requirement for life?

A Water B Oxygen C Energy D Plastic

1.2 Plants that produce their seeds in cones are …

A ferns. B mosses. C conifers. D flowering plants.

1.3 In a flowering plant, the male sex cells are found in the …

A ovules. B pollen grains. C stigma. D petals.

1.4 Which of these is a property of most metals?

A Brittle B Poor conductor C Malleable D Dull

1.5 The best method to separate a mixture of alcohol and water is …

A sieving. B distillation. C using a magnet. D filtration.

1.6 A solution with a pH of 7 is …

A acidic. B basic. C neutral. D a strong acid.

1.7 A vacuum flask keeps a drink hot mainly because …

A the vacuum stops heat transfer by conduction and convection. B the vacuum stops all radiation.

C the glass heats up the drink. D the lid produces heat.

1.8 The SI unit of energy is the …

A watt. B newton. C joule. D volt.

1.9 The Moon takes about … to orbit the Earth once.

A 24 hours B 7 days C 27 days D 365 days

1.10 When it is summer in South Africa, it is … in the Northern Hemisphere.

A summer B winter C spring D autumn

(10)

QUESTION 2: MATCHING COLUMNS

Choose a description from COLUMN B that matches the term in COLUMN A. Write only the letter (A–L) next to the question number, e.g. 2.11 M. Each letter may be used only once.

COLUMN A	COLUMN B
2.1 Lithosphere	A A substance that dissolves in a solvent
2.2 Hydrosphere	B An object that moves in an orbit around a planet
2.3 Atmosphere	C All the water on, under and above the Earth's surface
2.4 Germination	D The reaction of an acid with a base
2.5 Filtrate	E The layer of gases around the Earth
2.6 Solute	F An energy source that will run out one day
2.7 Neutralisation	G The liquid that passes through filter paper
2.8 Convection	H The solid, rocky outer layer of the Earth
2.9 Non-renewable	I When a seed starts to grow
2.10 Satellite	J Heat transfer by the movement of a liquid or a gas
	K A substance that does not dissolve
	L Energy that comes from the Sun

(10)

TOTAL SECTION A: [20]

SECTION B

QUESTION 3: LIFE AND LIVING

3.1 The biosphere overlaps with the lithosphere, the hydrosphere and the atmosphere. Explain what this means. (2)

3.2 Study this list of plants: maize, fern, pine tree, moss, rose

3.2.1 Which THREE plants are non-flowering plants? (3)

3.2.2 Maize is a monocotyledon and a rose is a dicotyledon. Give TWO differences between monocotyledons and dicotyledons. (4)

3.3 Spiders and insects are both arthropods. Give TWO ways in which a spider differs from an insect. (2)

3.4 (a) What is menstruation? (b) Name the male organ that produces sperm cells. (3)

3.5 (a) Give TWO examples of variation among learners in your class. (b) Why is variation within a species important for its survival? (4)

3.6 Name TWO conditions that a seed needs to germinate. (2)

[20]

QUESTION 4: MATTER AND MATERIALS

4.1 Choose the best material from the list for each object and give a property that makes it suitable: rubber, copper, glass, steel

4.1.1 Electrical wires (2)

4.1.2 A window pane (2)

4.1.3 A car tyre (2)

4.2 Give TWO reasons why it is important to recycle materials such as paper, glass and plastic. (2)

4.3 Describe how pure water can be obtained from salt water by distillation. (4)

4.4 (a) Name TWO acids found in the home. (b) Name TWO bases found in the home. (c) What colour does red cabbage indicator turn in an acid? (5)

4.5 (a) Sort these elements into metals and non-metals: magnesium, sulfur, aluminium, carbon. (b) Write down the chemical symbol for magnesium. (3)

[20]

QUESTION 5: ENERGY AND CHANGE

5.1 A learner drops a ball from a height of 2 m. Describe the energy changes from the moment the ball is let go until it hits the ground. (3)

5.2 On a roller coaster ride, where does the car have (a) the most gravitational potential energy and (b) the most kinetic energy? (2)

5.3 Grade 7 learners wrapped three identical tins of hot water in different materials. Each tin contained 200 mℓ of water at 80 °C. They measured the temperature after 20 minutes.

Tin	Starting temperature (°C)	Temperature after 20 min (°C)
1: no cover	80	52
2: cotton cloth	80	60
3: woollen scarf	80	68

5.3.1 Write down the independent variable in this investigation. (1)

5.3.2 Calculate the drop in temperature of tin 2. (1)

5.3.3 Which material is the best insulator? Give a reason from the results. (2)

5.3.4 Name TWO variables that were kept the same to make this a fair test. (2)

5.4 Name TWO renewable energy sources used to produce electricity in South Africa and give ONE advantage of each. (4)

5.5 (a) Why is electricity sent through the national grid at a very high voltage? (b) What does a step-down transformer near your home do? (3)

5.6 Give TWO ways in which a family can save electricity at home. (2)

[20]

QUESTION 6: PLANET EARTH AND BEYOND

6.1	Draw a labelled diagram to show the positions of the Sun, the Earth and the Moon at full moon.	(3)
6.2	Explain what happens during a lunar eclipse.	(2)
6.3	When do spring tides happen, and how are they different from other tides?	(2)
6.4	Give TWO differences between summer and winter in South Africa.	(2)
6.5	Explain why we have day and night.	(2)
6.6	Give TWO ways in which people in the past used the Moon and the stars in their daily lives.	(2)
6.7	Give TWO reasons why the Southern African Large Telescope (SALT) was built near Sutherland in the Karoo.	(2)
6.8	Explain why the Moon seems to change shape during the month, even though it stays round.	(3)
6.9	Why do we have a leap year (366 days) every four years?	(2)

[20]

TOTAL SECTION B: [80]

TOTAL: 100

END OF QUESTION PAPER